Where Does Vision Belong in a Frozen Time-Series Forecaster?

A placement measurement for satellite-conditioned photovoltaic forecasting

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Abstract. Adding a second modality to a forecaster is easy; establishing that the forecaster **uses** it is not. This work takes a frozen time-series foundation model and a frozen self-supervised video encoder and asks a deliberately narrow question: given that both networks are fixed, **where** in the forecasting pipeline must the visual signal be attached before the numeric model actually consults it? Three attachment points spanning the standard fusion taxonomy were implemented, trained under an identical recipe, and evaluated with a counterfactual instrument that re-runs the test set with the imagery withheld, so that the quantity reported is the model's **reliance** on vision rather than its overall accuracy. Two of the three attachments — pooled late fusion and mid-sequence token injection — produce a measured reliance statistically indistinguishable from zero against a decision threshold fixed before the runs. The third, in which each forecast position issues its own cross-attention query against an unpooled visual memory, produces a reliance five times that threshold, reproducibly across three seeds, and lifts the headline skill score from 0.523 to 0.547. The contribution is therefore not a new attention mechanism but a controlled placement result with two informative negatives, together with a reusable protocol for proving that a second modality is doing work.

1 Settings

All metrics are normalised by installed capacity, so they are comparable across systems of different size, and all are computed on daytime steps only.

Quantity / Metric	Specification / Description
Installations	100 UK residential rooftops, 1.5–4.0 kW; two carry a data-quality flag and are dropped, leaving 98
Sampling	every 30 minutes, years 2019 and 2020
History window	14 days = 672 samples of the plant's own power
Forecast horizon	6 hours = 12 samples, 9 quantiles emitted per step
Covariates	14 auxiliary channels known in advance for the future window as well as the past
Visual signal	8 satellite frames at 224 × 224 covering the 6 hours ending at the origin, centred on the installation
Split	70/15/15 **by installation**, not by time: train, validation and test share the same two years and share no rooftop
Evaluation set	14 test installations, 165,295 scored daytime steps
Seeds	42, 43, 44
Skill score (SS, ↑)	$\mathrm{SS} = 1 - \mathrm{NRMSE}_{\mathrm{model}}/\mathrm{NRMSE}_{\mathrm{ref}}$ against Smart Persistence ($\mathrm{SS} = 0$). Higher is better; one means perfection.
Ramp NMAE (↓)	NMAE restricted to steps where true power changes sharply between consecutive samples — a cloud edge crossing the array (lower is better).
Point / Distr. metrics	NMAE (↓), NRMSE (↓), CRPS (↓, proper scoring rule across 9 quantiles; collapses to MAE for point forecasts).

Table 1: The experimental setting and evaluated metrics. All configurations in this report share this setting; the only variable across arms is where the visual signal attaches. Primary reported metrics are Skill score (relative to Smart Persistence) and Ramp NMAE (restricted to high-variability cloud transitions).

Smart Persistence is the field's standard null hypothesis and it is a strong one: on clear or uniformly overcast days it is nearly unbeatable.

Ramps deserve the same emphasis. They are a small minority of steps, but they carry nearly all of the operational cost of a forecast error.

2 The research question

The multimodal forecasting literature is largely a literature of proposals: an architecture is introduced, it is trained end to end, it beats a set of unimodal baselines, and the improvement is attributed to the additional modality. That attribution is rarely tested. When both encoders are trained jointly, an accuracy gain is consistent with at least three explanations — the model learned to read the images, the extra capacity helped, or the extra gradient path acted as regularisation.

This work inverts the emphasis. Both encoders are **pretrained and frozen**, so neither can adapt to the other; only a small bridge between them is learned. The architecture is then held fixed and the **attachment point** is varied. The question becomes:

> *Given a frozen sequence forecaster and a frozen visual encoder, at which point in the forecaster's computation must the visual representation be introduced before the forecaster demonstrably uses it?*

Freezing is what makes the question answerable. With both backbones fixed, no configuration can win by having more capacity than another, and any measured difference is attributable to the wiring. It is also what makes the result portable: the finding is a statement about placement in a frozen pipeline, not about one particular set of weights.

3 Components

Component	Role and dimensions	State
Chronos-2 [1]	Pretrained encoder-only time-series transformer.	frozen, top 3 of 12 blocks unfrozen
V-JEPA 2 [2]	Self-supervised video encoder. Encodes the 8-frame clip once, offline, into 4 temporal slices × 196 spatial patches × 1024 channels, cached.	frozen entirely
Channel projection	Maps the visual channel width onto the backbone's width.	trained
Fusion module	The only thing that differs between configurations.	trained

Table 2: The two pretrained networks and the learned bridge between them. Everything of consequence is frozen; the trainable budget is the bridge plus the top quarter of the forecasting encoder.

3.1 The forecasting backbone

The backbone treats a series the way a transformer treats any sequence: it is cut into non-overlapping patches of 16 consecutive samples, each patch is linearly projected into a token, and a 672-step history becomes 42 context tokens. Future positions are represented by additional placeholder tokens carrying only their known covariates; self-attention over the whole assembly lets those future positions absorb information from the past, and a projection head reads the answer off them.

The backbone is genuinely general-purpose: it was pretrained on heterogeneous series and has no notion of irradiance, panel tilt or clear-sky curves, so nothing about the solar domain is baked in. It also consumes covariates natively, which means the numeric-only control is already a strong model and the visual signal must earn its place against a high bar.

3.2 The visual encoder

The choice of a self-supervised video model rather than a caption-aligned one is deliberate. The relevant visual content — the motion and deformation of cloud fields — is temporal and has no useful textual description; a representation trained to match captions would discard exactly the structure that matters. V-JEPA's predictive objective in latent space is, by construction, a model of **how a scene changes**, which is the property the forecaster needs. The encoder is never run during training.

4 The four configurations

The three fusion arms are best understood not as three architectures but as three answers to a single structural question: **along which axis does the visual information enter?**

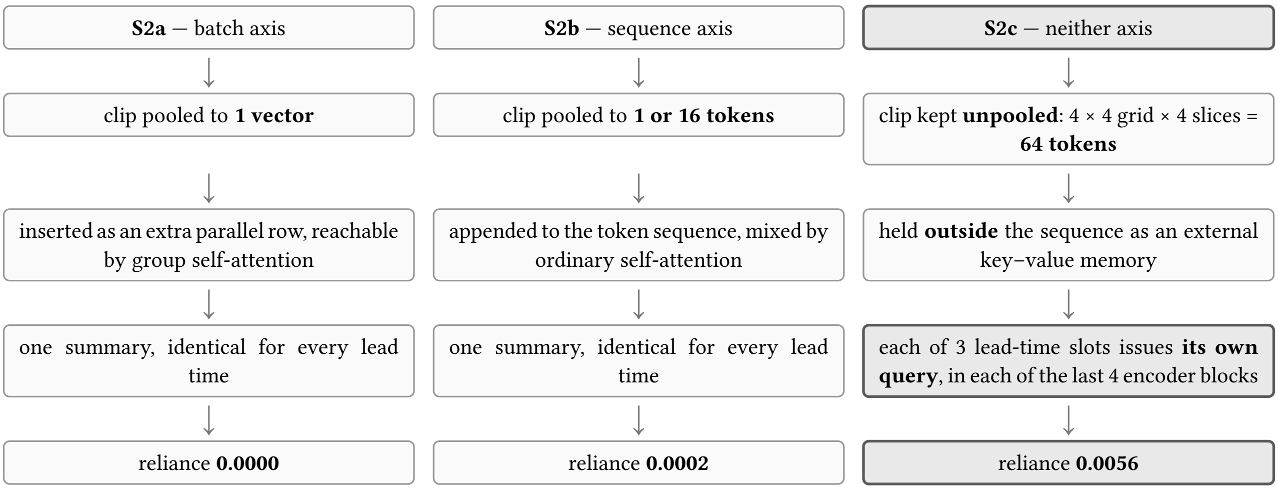

Figure 1: The fusion taxonomy as a single structural question. The two arms that summarise the imagery before the forecaster has an opinion measure zero reliance; the arm that refuses to summarise does not.

4.1 S1 — the vision-free control

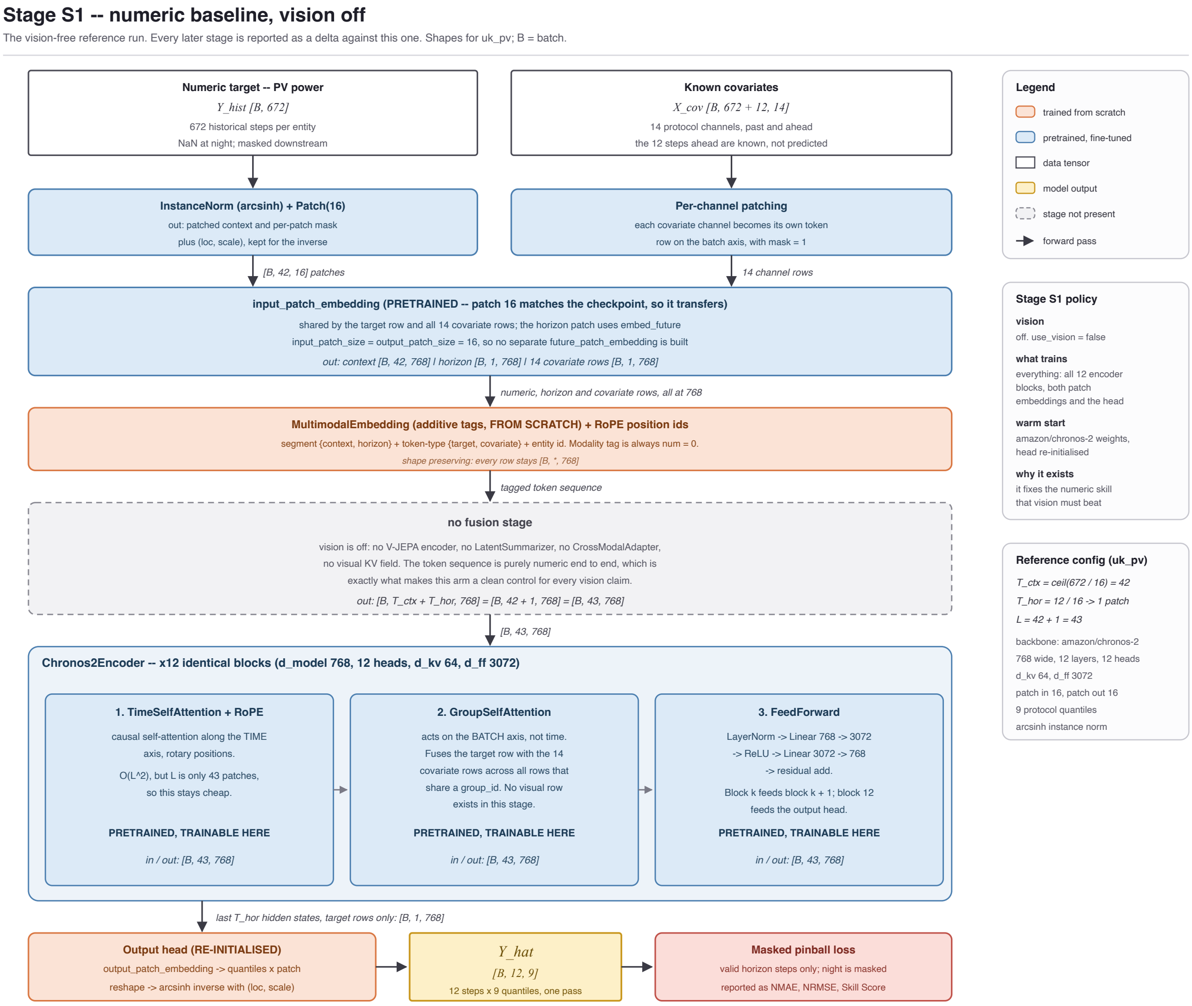

Figure 2: S1, the numeric-only control. History patching, covariate rows and the frozen backbone, with no visual pathway. Every fusion arm is warm-started from this checkpoint, so it defines the zero of the reliance measurement.

S1 is the same backbone with no imagery at all. It exists to define the baseline against which reliance is measured, ensuring that a multimodal model's advantage came from the fine-tuning recipe rather than from the second modality. Every subsequent arm is initialised from this checkpoint and trained with the same schedule, so any difference is attributable to the visual pathway.

4.2 S2a — fusion on the batch axis

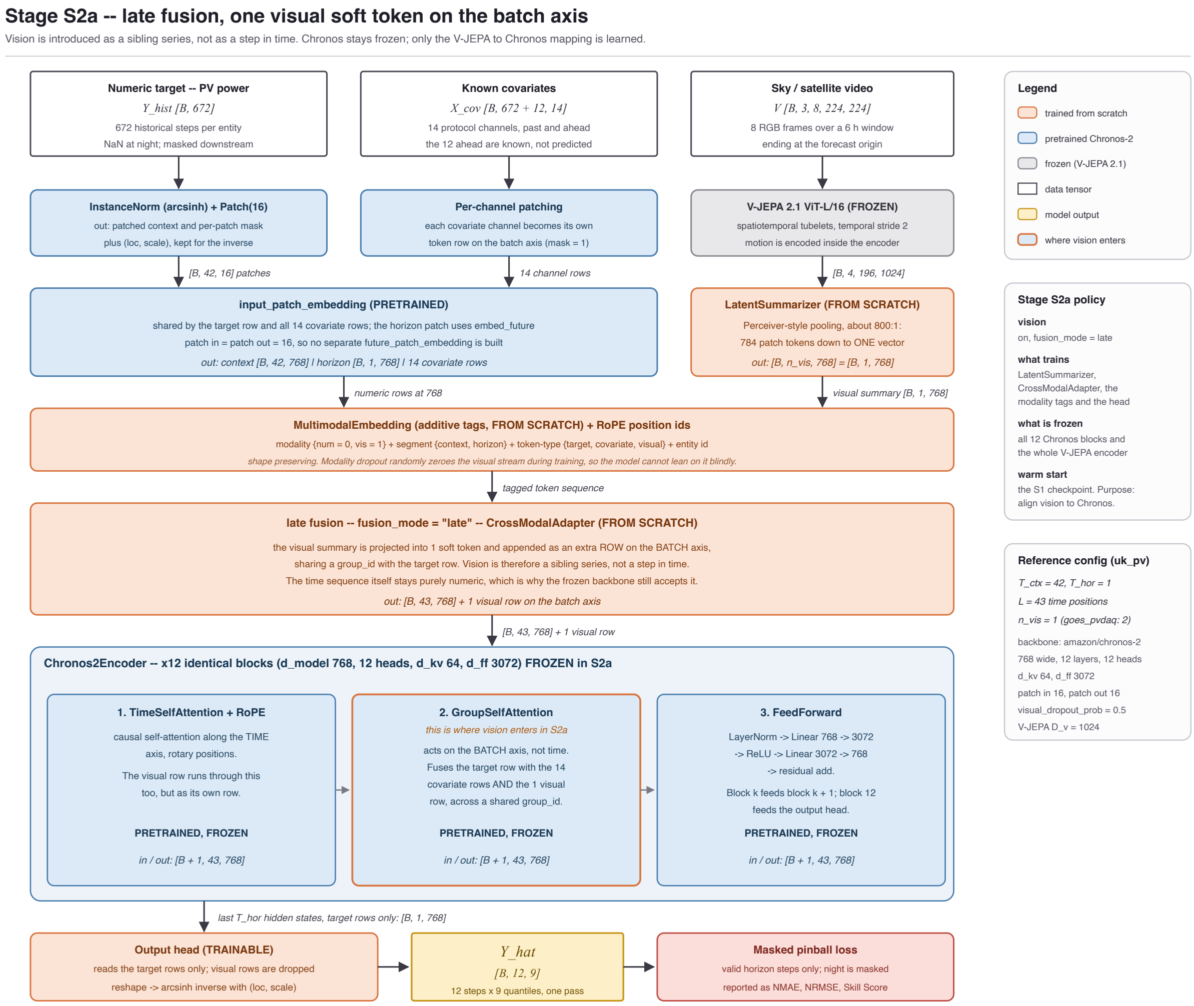

Figure 3: S2a, pooled late fusion. The clip is compressed to a single vector and inserted as an additional row on the batch axis, where the backbone's group self-attention can reach it. The visual content is summarised before the forecaster is consulted.

The clip is pooled to a single descriptor and presented to the backbone as an extra parallel channel — a summary of "what today's sky is like", available to every forecast position equally. This is textbook late fusion, and it was the first arm precisely **because** it is the textbook answer: if the standard method works, the thesis is finished, and if it does not, the null is a result rather than an omission.

The rationale for expecting it to work is reasonable. The most obvious use of a satellite view is as a coarse regime indicator — clear, broken, overcast — and a single vector is an adequate carrier for a regime label. The rationale for expecting it to fail is equally clear in hindsight: the pooling operation compresses roughly 800 visual elements into 1, and it does so **before anything has been asked of them**. The summary must be computed without knowing which part of the scene is relevant, and by the time the forecaster is in a position to have an opinion, the discarded detail is gone.

4.3 S2b — fusion on the sequence axis

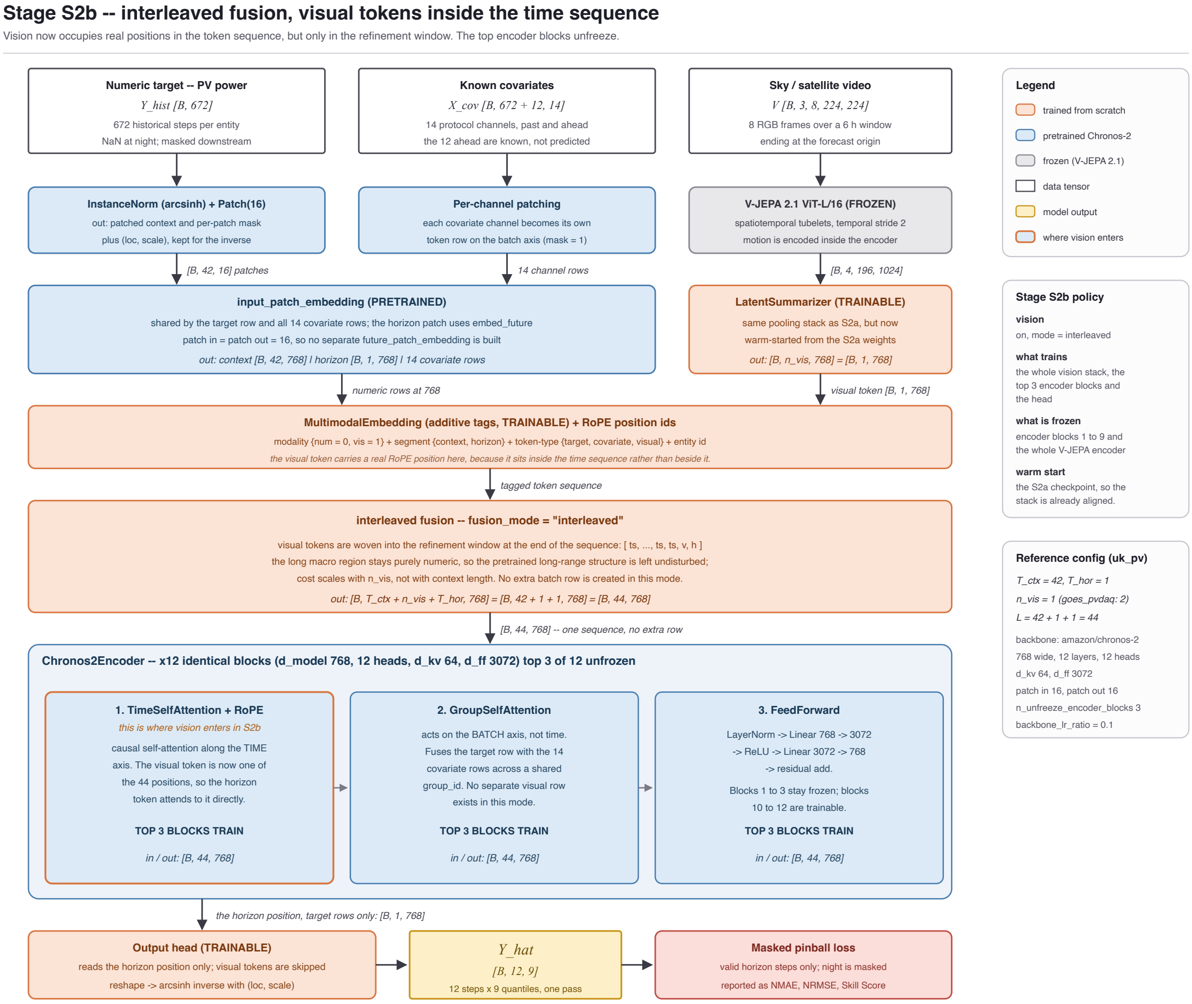

Figure 4: S2b, mid-sequence injection. Visual tokens are appended to the token sequence itself, so ordinary self-attention mixes them with the numeric history. Shown at the single-token setting; a wider variant with 16 tokens was also run.

The natural next move is to stop treating vision as a parallel channel and make it part of the sequence, so that ordinary self-attention can relate visual tokens to numeric ones directly. Two widths were run — one visual token, and sixteen — to test whether the failure of S2a was simply a matter of bandwidth.

The reasoning is sound and the result is instructive. Widening the channel from 1 token to 16 did not help; the reliance stayed at zero within noise in both settings. That comparison is what rules out the bandwidth explanation and points at something structural: the problem is not **how much** visual information is admitted but **when** it is summarised relative to when it is needed. In both S2a and S2b the imagery is condensed into a fixed representation before the model has formed any view about the forecast, and every forecast position then receives the same representation.

4.4 S2c — the forecast queries the sky

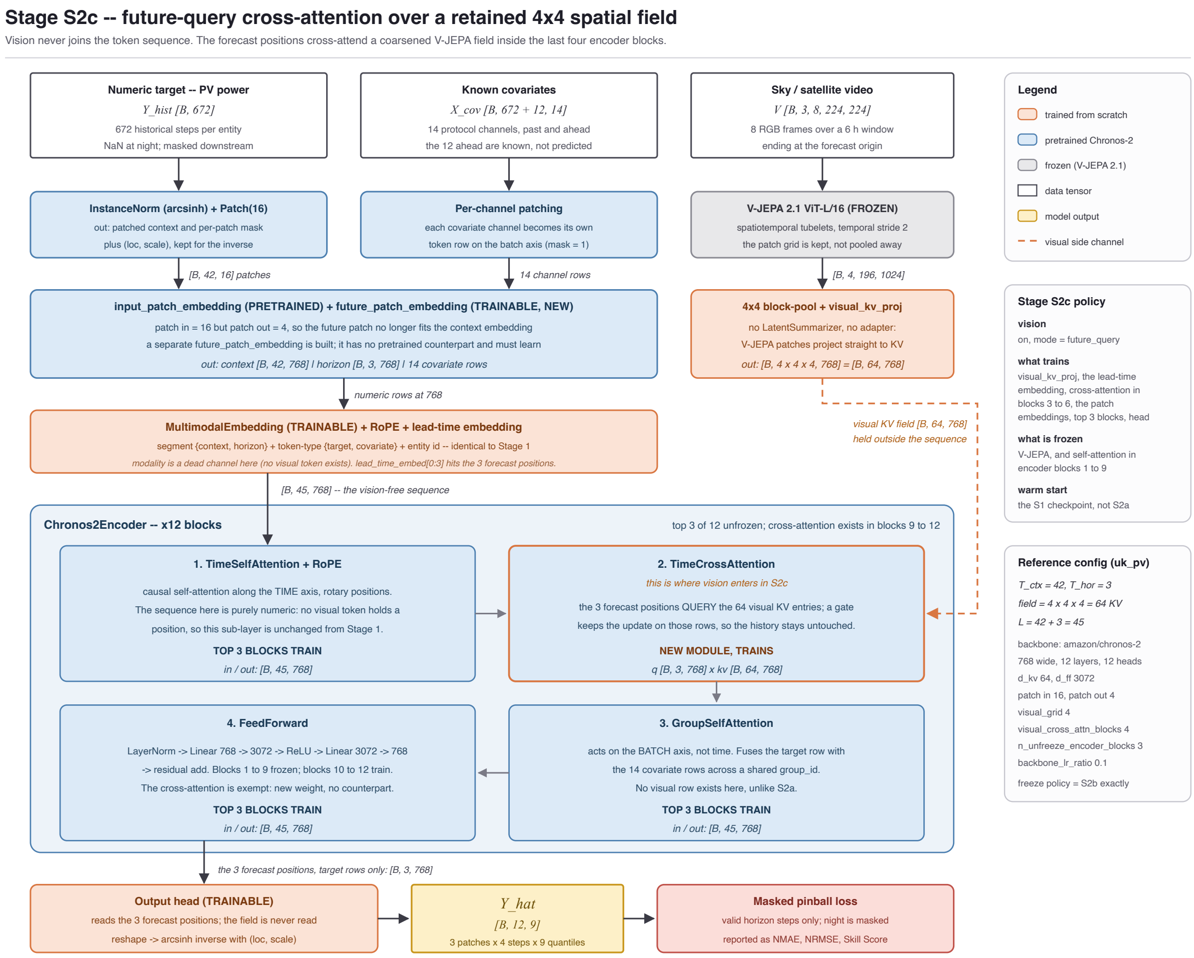

Figure 5: S2c, future-position cross-attention. The visual field is retained unpooled as a 64-token external memory, outside the sequence. Each forecast position issues its own query against it, in each of the backbone's last four encoder blocks.

S2c changes the direction of the operation. The imagery is not inserted anywhere. It is retained **unpooled** as an external memory — the 14 × 14 patch grid is block-pooled to 4 × 4 and all four temporal slices are kept, giving 64 key–value tokens — and the forecast horizon is subdivided into three lead-time slots. Each slot issues its own cross-attention query against that memory, in each of the backbone's last four encoder blocks. No summarisation happens anywhere: the pooled descriptor and the projection adapter used by the other arms are both bypassed.

4.4.1 What a query actually is

A query is not made of the thing that is being predicted. It is a **description of what is being looked for** — the same object as a search string, which exists before the document it retrieves. Each forecast slot's query is assembled from exactly two ingredients, both available before any forecast value exists: a learned lead-time identity, a vector attached permanently to "the slot two hours out", shared across all samples and installations and fixed after training; and the slot's own hidden state after self-attention over the history and the known covariates, which encodes a **belief** about what is likely to happen, not the outcome.

self-attention populates the slot from history and covariates → slot forms a **query**: lead-time identity + its own belief → query reads the **64 unpooled visual tokens** → result **gated back** onto that slot alone → projection head emits the number

Figure 6: Order of operations for one lead-time slot in S2c, repeated in each of the last four encoder blocks. Each pass refines the slot's question with what the previous round returned.

4.4.2 *Why the slots ask different things*

Each lead-time slot receives gradient only from its own horizon's errors. The two-hour slot is penalised for two-hour mistakes, the five-hour slot for five-hour mistakes, and cloud fields relevant at those two horizons sit at different distances from the array. Nothing in the design instructs a slot to attend to cloud edges; the differentiation is the accumulated residue of one horizon's own past errors.

4.4.3 *Why this is not late fusion, and not another cross-attention paper*

"This is late fusion with extra steps." Late fusion is in this work — it is S2a, it was built properly, and it is one of the measured negatives. The distinguishing test is the one stated in the box above: in S2a and S2b one can point at the sky representation before the forecast is made; in S2c there is nothing to point at. What exists at that moment is 64 uncombined visual elements and three questions not yet asked. The visual contribution is **manufactured on demand**, once per lead time, and comes out different each time. A model holding a single summary cannot do this even in principle.

"Cross-attention between a forecaster and satellite imagery already exists."

	CrossViViT [3]	**S2TX [4]**	**This work**
What the attention joins	numeric history ↔ satellite context	long-range state space ↔ short local window	**numeric forecast ↔ satellite imagery**
Modalities	two	one (numeric only)	two
Queries issued from	**history** timesteps, one per past step, folded into the batch dimension	encoder positions over the history	**future** positions, one per lead-time slot
Forecast produced	afterwards, by a separate temporal module	afterwards, by a prediction head	**the fused state is the forecast state** — no separate head consumes it
Positional encoding on keys	2-D geographic	temporal	none — see Section 8
Training	both encoders from scratch	end to end from scratch	both backbones frozen; only the bridge is learned
Kind of claim	an architecture, scored on benchmarks	an architecture, scored on benchmarks	**a placement measurement, with two nulls**
Is alignment hard?	yes	no — every variate shares one clock and one table	yes — that difficulty is the subject

Table 3: The same operator, three different uses. The contribution claimed here is not the attention mechanism — standard plumbing in all three columns — but the demonstration that its **placement** determines whether the modality is used at all, supported by two controlled negatives that a proposal-style paper would not have built.

The last row separates the two settings. Adding a satellite view is not the same kind of act as adding another numeric channel: cross-variate reasoning operates over columns of one aligned table, where every variate has a value at every timestep, on the same clock, in comparable units. A sky observation is none of those things. The entire difficulty this work measures — that two of three plausible wirings deliver exactly nothing — does not arise in the cross-variate setting at all, because there the alignment is free.

5 The measurement instrument

After training, the model is frozen and the test set is evaluated twice: once normally, and once with the visual query switched off. The difference between the two passes is the **reliance** — the amount of the model's accuracy that depends on it having seen the sky.

This is a stronger instrument than comparing a multimodal model to a unimodal one, because it holds every weight fixed. It cannot be confounded by capacity, initialisation, or training schedule; the two passes differ only in whether the imagery was available.

6 Results

6.1 The placement result

Table 4 is the central finding. The reliance column reports the ramp-error improvement attributable to the imagery, measured by the counterfactual pass.

Arm	Where vision enters	Reliance (ramp)
S2a	batch axis, pooled	0.0000 ± 0.0015
S2b	sequence axis, 1 token	0.0006
S2b wide	sequence axis, 16 tokens	0.0002 ± 0.0016
S2c	**future-position queries**	**0.0056 ± 0.0006**

Table 4: The placement ladder.

6.2 Controls and ablations

Two perturbation experiments test whether S2c's gain reflects genuine visual content.

Control	Perturbation	NMAE	Skill	Reliance (ramp)
S2c	none	0.0700 ± 0.0012	0.5470 ± 0.0060	+0.0056 ± 0.0006
A09	temporal order of frames shuffled	0.0700 ± 0.0012	0.5470 ± 0.0060	+0.0056 ± 0.0006
A10	imagery swapped with another plant's	0.0901 ± 0.0023	0.4222 ± 0.0148	−0.0082 ± 0.0016

Table 5: Perturbation controls on S2c. Frame shuffling has no effect, as predicted by the absence of positional encoding on the visual memory. Substituting another installation's imagery destroys the gain and inverts the reliance.

A09 returns results bit-identical to the unperturbed model at every digit and every seed. This is not a null result to be explained away; it is a **confirmed structural prediction**. The visual memory carries no positional encoding — a design consequence discussed as a limitation in Section 8 — so the 64 tokens form an unordered set and shuffling their temporal order is provably a no-op. The experiment verifies that the implementation matches the analysis.

A10 feeds the model a different installation's sky. NMAE degrades from 0.0700 to 0.0901 — worse than the vision-free control — and the reliance turns **negative**: with mismatched imagery, vision now actively harms the forecast. A model that had learned a generic "cloudy day" prior would be roughly indifferent to whose sky it was shown. This one is not. The gain is therefore attributable to plant-specific visual content, which is the strongest available evidence short of an interpretability study.

> **Reporting note.** The A10 result files are present and dated at the current revision, while the internal experiment registry still lists the run as configured-but-not-executed. The registry entry is stale rather than the result being provisional, but the discrepancy is flagged here rather than silently reconciled.

A separate diagnostic supports the design decision to keep the visual field unpooled. A linear probe was trained to predict short-horizon power change from the visual representation alone (Table 6). The information that pooling destroys is measurable, it lives in the spatial layout, and its temporal reach matches the horizon over which the fusion arm helps.

Visual resolution	+30 min	+60 min	+120 min
1 × 1 (pooled descriptor, as in S2a)	0.006	—	—
4 × 4 (as used by S2c)	**0.051**	**0.082**	≈ 0

Table 6: Linear probe on the visual representation alone: variance explained (R^2) in short-horizon power change. Pooling to a single descriptor removes almost all of the recoverable signal.

6.3 Placement against the baseline suite

Rank	Model	Second modality	Where the fusion sits	Skill score	Ramp NMAE
			This work (MMTSFM placement arms)		
1	**MMTSFM S2c** (ours)	satellite	cross-attention from **future** positions to unpooled memory	**0.5470**	0.1461
3	MMTSFM S2b wide (ours)	satellite	sequence axis, 16 appended tokens (self-attention)	0.5352	0.1484
4	MMTSFM S2b (ours)	satellite	sequence axis, 1 appended token (self-attention)	0.5322	0.1487
5	MMTSFM S2a (ours)	satellite	batch axis, pooled vector (group self-attention)	0.5258	0.1487
7	MMTSFM S1 control (ours)	none	— (vision-free control)	0.5230	0.1506
			Multimodal forecasters (satellite & pseudo-image)		
2	Time-VLM [5]	series **rendered as** images	reuses visual pretraining in the opposite direction	0.5404	—
13	Solar-VLM [6]	satellite + text	vision–language fusion, multi-site	0.4396	0.1514
21	CrossViViT [3]	satellite	cross-attention from history timesteps	0.3491	—
26	Aurora [7]	several	joint multimodal pretraining	0.2324	—
27	SUNSET [8]	sky/satellite	convolutional precedent; joint encoding	0.2162	—
28	UniCast [9]	several	prompting a foundation forecaster	0.1211	—
30	VisionTS++ [10]	series **rendered as** images	continual pretraining of a visual backbone	0.0167	—

Table 7: Multimodal leaderboard: MMTSFM placement arms and prior multimodal forecasters, indexed by overall rank on Skill score. Skill is relative to Smart Persistence. Dashes indicate unrecorded ramp subsets or absence of a second modality / fusion mechanism.

Rank	Model	Second modality	Where the fusion sits	Skill score	Ramp NMAE
		Supervised deep learning			
6	iTransformer + covariates	covariates	channel-inverted self-attention over variates	0.5257	**0.1445**
12	PatchTST	none	—	0.4581	0.1543
14	Temporal Fusion Transformer	covariates	gated residual & temporal self-attention	0.4264	0.1605
15	MLP	none	—	0.4219	0.1624
22	DLinear	none	—	0.3231	0.1746
		Retrieval & frozen-backbone adaptation			
9	TS-RAG [11]	retrieved **numeric** history	concatenated to the context	0.4779	—
10	Cross-RAG [12]	retrieved **numeric** history	cross-attention between query and retrievals	0.4768	—
18	CoRA [13]	covariates	residual adapter on frozen backbone	0.3798	0.1624
		Time-series foundation models (zero-shot & fine-tuned)			
8	Chronos-2, fine-tuned	covariates	group self-attention, fine-tuned	0.5042	0.1494
11	Chronos-2, zero-shot	covariates	group self-attention, zero-shot	0.4737	0.1544
20	TTM, fine-tuned	covariates	MLP-Mixer temporal/ channel mixing, fine-tuned	0.3601	0.1716
23	TiRex, zero-shot	none	—	0.2873	0.1826
24	TimesFM, zero-shot	none	—	0.2708	0.1902
32	TTM, zero-shot	covariates	MLP-Mixer, zero-shot	−0.0807	0.2922

Table 8: Unimodal deep learning and foundation model baselines: supervised architectures, retrieval/adapters, and zero-shot/fine-tuned TSFMs, indexed by overall rank on Skill score.

Rank	Model	Second modality	Where the fusion sits	Skill score	Ramp NMAE
			Tabular models & classical ML		
16	TabPFN	none	—	0.4063	0.1631
17	LightGBM	covariates	gradient-boosted trees over tabular lags	0.3854	0.1672
19	TabFM ensemble	none	—	0.3626	0.1573
			Reference & statistical baselines		
25	Hourly climatology	none	—	0.2337	0.1665
29	Seasonal naive	none	—	0.1068	0.2056
31	Persistence	none	—	0.0141	0.2550

Table 9: Tabular, classical ML, and reference baselines: tree models, tabular foundation models, and statistical persistence/climatology floors, indexed by overall rank on Skill score.

7 Related work and positioning

Work	Second modality	Where the fusion sits	Encoders
Multimodal PV and irradiance forecasting			
CrossViViT [3]	satellite	cross-attention from history timesteps, geographic positional encodings	trained
SUNSET [8]	sky/satellite	convolutional precedent; joint encoding	trained
PV-VLM [14]	sky images + text	vision–language fusion through a large pretrained model	partly frozen
M3S-Net [15]	imagery	multi-scale feature fusion	trained
Solar-VLM [6]	satellite + text	vision–language fusion, multi-site	partly frozen
SPIRIT [16]	sky camera	frozen visual features + physics-inspired descriptors	frozen
General multimodal and cross-modal forecasting			
UniCast [9]	several	prompting a foundation forecaster	frozen backbone
Aurora [7]	several	joint multimodal pretraining	trained
Time-VLM [5]	series **rendered as** images	reuses visual pretraining in the opposite direction	frozen backbone
VisionTS++ [10]	series **rendered as** images	continual pretraining of a visual backbone	trained
S2TX [4]	none — numeric only	cross-attention between two time scales, inside the encoder	trained
Adapting a frozen forecaster — the nearest methodological neighbours			
TS-RAG [11]	retrieved **numeric** history	concatenated to the context	frozen backbone
Cross-RAG [12]	retrieved **numeric** history	cross-attention between query and retrievals	frozen backbone
CoRA [13]	covariates	wrapper over a frozen Chronos-2; in the leaderboard on identical data	frozen backbone
This work			
S2c	satellite	cross-attention from **future** positions to an unpooled external memory	both frozen

Table 10: Positioning. Every entry in the first two groups is a **proposal**: an architecture is introduced and accuracy is demonstrated. None isolates the placement variable with the modality held fixed, and none reports a counterfactual reliance measurement.

8 Limitations

Limitation	Consequence	Remedy
The visual memory is **permutation-invariant** — no positional encoding is applied to the 64 key–value tokens	Forecast slots address the field **by content**, not by location: the model can ask “is there a bright convective texture anywhere” but not “is it to the west”. None of the advection geometry that motivated the modality is currently exploited, so the measured reliance is a **lower bound**. It also makes the frame-shuffling control a prov-	Add rotary or learned spatial encodings to the visual keys — the single highest-value follow-up.
[illegible]	[illegible]	[illegible]

Table 11: Known limitations, their effect on the reported numbers, and the corresponding remedy. The first is the most consequential and its fix is already specified.

9 Scope of the claim

The mechanism is domain-agnostic; the evidence is not. Nothing in S2c encodes solar physics — the visual encoder was trained on ordinary internet video, the forecasting backbone on generic series, and the bridge contains no daylight mask, clear-sky prior or geographic term. The honest description of the architecture is: **a frozen sequence forecaster that can, at each position of its own forecast, query an unreduced memory produced by a frozen encoder of another modality.**

It is equally not a general-purpose forecasting architecture. Given a univariate benchmark with no side information it has nothing to attend to and degenerates to the plain forecaster. It is a **conditioning mechanism**, and Table 12 states the conditions under which it should be expected to pay off — which is itself part of the contribution, because it tells the next reader when to reach for this and when not to bother.

Precondition	Traffic flow	Clinical deterioration	Predictive maintenance
Side information is high-dimensional and unstructured	upstream road-camera or overhead views	imaging and free-text notes	vibration or acoustic spectrograms
Which part matters varies per sample	congestion location changes hourly	the relevant finding differs per patient	the failing component differs per machine
Different horizons care about different parts	congestion three junctions away matters at 40 min, not at 5	a pending culture bears on tomorrow, not the next hour	an early bearing signature predicts failure at a horizon, not now
Signal is causally upstream with a lag	traffic propagates along the network	physiology deteriorates on a delay	wear precedes failure

Table 12: Preconditions for transfer, and three domains that satisfy them. PV satisfies all four, which is why it is a good testbed — not because the work is about solar energy.

The traffic case is the sharpest comparison, because it is the one where the known weakness would bite hardest: **which** junction the congestion sits on is most of the information, and the visual memory deliberately carries no position. Transfer there would very likely require the spatial- encoding fix of Table 11 first.

The most portable element is arguably not the module but the protocol: freeze the model, run the evaluation twice, withhold one modality, and compare against a threshold fixed in advance. That procedure is independent of solar, of imagery, and of this architecture, and it applies to any multimodal forecaster.

10 Summary

1. Two of the three standard placements for a visual modality in a frozen forecaster — pooled late fusion and mid-sequence injection — deliver **no measurable reliance** on that modality, with tight central estimates of 0.0000 and 0.0002 against a pre-registered threshold of 0.0011.
2. Widening the injected representation sixteenfold does not change this, which excludes bandwidth as the explanation and localises the failure to **when the imagery is summarised** relative to when it is needed.
3. Attaching the modality as an unpooled external memory queried per forecast position yields a reliance of 0.0056 ± 0.0006 — five times the threshold, on every seed — and raises the skill score over Smart Persistence from 0.523 to 0.547.
4. Substituting another installation's imagery destroys the gain and inverts the reliance, establishing that the model reads plant-specific visual content rather than a generic weather prior.
5. The visual memory currently carries no positional information, so the geometric content of the scene is unexploited. The reported effect is a lower bound, and the corresponding experiment is already specified.

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